Question 7.4.7

AI25BTECH11040 - Vivaan Parashar

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1 Question:

The intercept of the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$ is AB. Equation of the circle with AB as a diameter is __________.

2 Solution:

Let's assume the two points, $\mathbf{A}$ and $\mathbf{B}$, where the line $y = x$ intersects the circle $x^2 + y^2 - 2x = 0$. The two points lie on the line, whose equation can be expressed in matrix form as:

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}^{\mathrm{T}} \mathbf{x} = 0 \tag{1}$$

Therefore, we have:

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \mathbf{A} = 0 \tag{2}$$

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \mathbf{B} = 0 \tag{3}$$

$$\implies \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix} \tag{4}$$

We also have the equation of the circle in the matrix form as:

$$\|\mathbf{x}\|^2 + 2\mathbf{u}^{\mathrm{T}}\mathbf{x} + f = 0 \tag{5}$$

$$\text{where, } \mathbf{u} = -\mathbf{O}, f = \|\mathbf{O}\|^2 - r^2 \tag{6}$$

Here, $\mathbf{O}$ is the center of the circle and r is its radius. Comparing the given equation of the circle ($(x-1)^2 + y^2 = 1$) with the above equation, we get:

$$\mathbf{O} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, r = 1 \tag{7}$$

Thus, we have the equation of the circle as:

$$\mathbf{u} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, f = 0 \tag{8}$$

$$\implies \|\mathbf{x}\|^2 - 2\begin{pmatrix} 1 & 0 \end{pmatrix}\mathbf{x} = 0 \tag{9}$$

Since both points **A** and **B** lie on the circle, we have:

$$\|\mathbf{A}\|^2 - 2\begin{pmatrix} 1 & 0 \end{pmatrix}\mathbf{A} = 0 \tag{10}$$

$$\|\mathbf{B}\|^2 - 2\begin{pmatrix} 1 & 0 \end{pmatrix}\mathbf{B} = 0 \tag{11}$$

$$\implies \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}^{\top}\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} - 2\begin{pmatrix} 1 & 0 \end{pmatrix}\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix} \tag{12}$$

3 Plot:

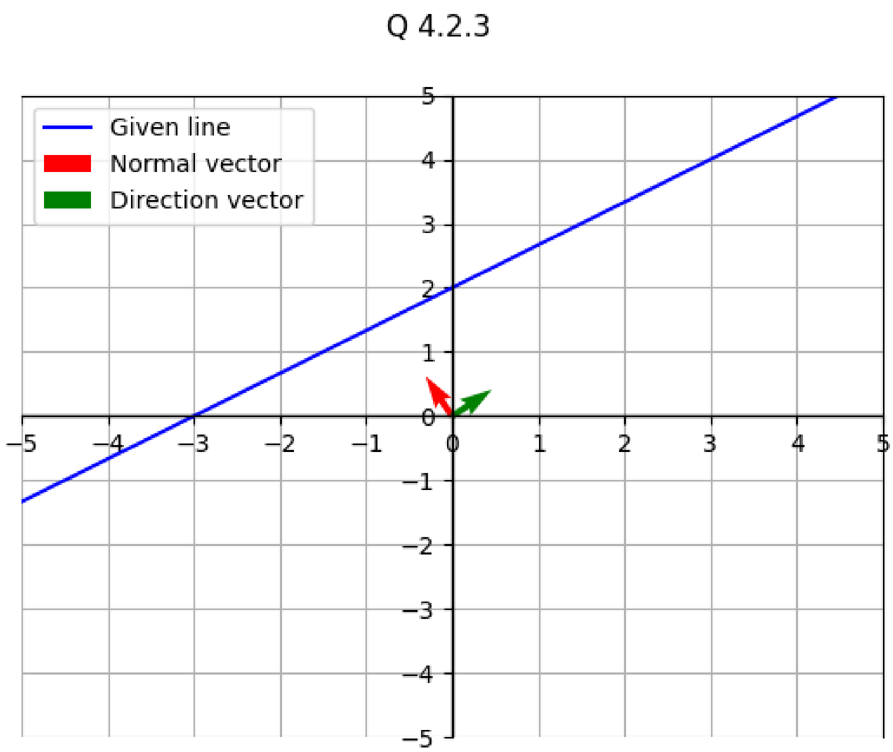

Figure 1: Graph of line with direction and normal vectors